

Programming 2025/2026

Laboratory

Session 1 (April 14-17)

For Loop and Range

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For Loop

```
for <var> in <seq>:  
    expression inside the for  
    ...  
    expression inside the for  
expression outside the for
```

- `seq` is a Python sequence
- `var` is a Python variable which holds a different value of `seq` (in the sequence order) at each iteration of the loop
- The expressions inside the for loop are repeated at each iteration of the loop (indentation!!!)
 - At the first iteration, `var` contains the first value of `seq`
 - At the last iteration, `var` contains the last value of `seq`
 - The iteration lasts as long as the size of `seq`

For Loop

Try the following piece of code and check the output

```
for character in "Andrea":  
    print(character)
```

```
for character in "Andrea":  
    print("Hello")
```

Exercise (4.1)

- Write a program that prints the number of times a given character appears in a given string

Exercise (4.2)

- Write a program that removes from a given string all the white spaces and the punctuation characters, and prints the resulting string
- Remember that the string module contains some useful constants

```
string.ascii_letters  
string.ascii_lowercase  
string.ascii_uppercase  
string.digits  
string.punctuation  
string.whitespace
```

Exercise (4.3)

- Write a program that given a string, prints the consonants and the number of vowels in the string while ignoring the white spaces, punctuation and other symbols

Range

- Generates a sequence of integers in a given range
 - It is an immutable sequence
- `range(a)`
 - Generates a sequence of integers in the range $[0, a)$, i.e. $0, 1, \dots, a-1$
- `range(a, b)`
 - Generates a sequence of integers in the range $[a, b)$, i.e. $a, a+1, \dots, b-1$
- `range(a, b, s)`
 - when $s > 0$, generates a sequence of integers $a, a + s, a + 2*s, \dots, a + i*s$ with $a + i*s < b$
 - when $s < 0$, generates a sequence of integers $a, a + s, a + 2*s, \dots, a + i*s$ but s being negative with $a + i*s > b$

Range

Try this code and check the output

```
for i in range(10):  
    print(i)  
  
for i in range(1,10):  
    print(i)  
  
for i in range(1,10,2):  
    print(i)  
  
for i in range(10,1,-2):  
    print(i)
```


Exercise (4.4)

- Write a program that prints the multiplication table of a given number n (for example from 0 to 10). For instance, given $n = 3$, the program prints:

$3 \times 1 = 3$	$3 \times 6 = 18$
$3 \times 2 = 6$	$3 \times 7 = 21$
$3 \times 3 = 9$	$3 \times 8 = 24$
$3 \times 4 = 12$	$3 \times 9 = 27$
$3 \times 5 = 15$	$3 \times 10 = 30$

Exercise (4.5)

- What about doing the multiplication table for all numbers between 1 and the given number n ?

Exercise (4.6)

- Write a program that checks if a given string is a palindrome or not using a for loop.

Exercise (4.7)

- Write a program that prints the greatest common divisor (gcd) of two given non-negative integers. For example, the gcd of 15 and 6 is 3.